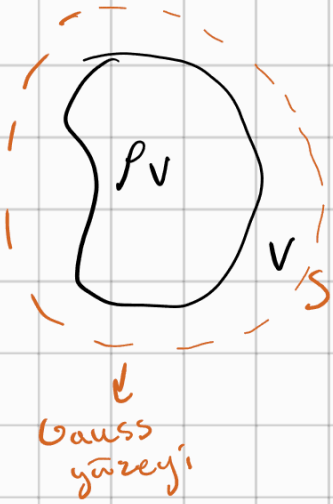


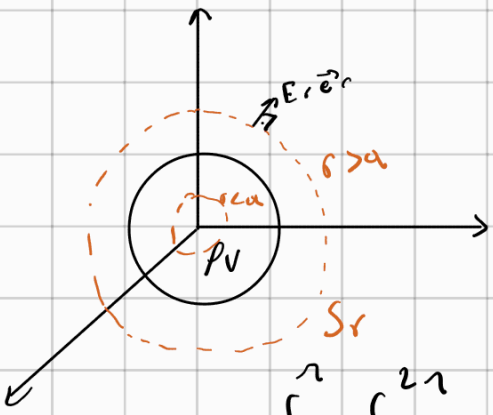
$$Q = \int_V \rho_V \, dV$$

$$\vec{D} = \varepsilon_0 \vec{E}$$

$$\int_S \vec{D} \cdot d\vec{s} = Q$$

S32 Pr 1

S17 Pr.1'i gauss ile çözünüz



$$\int_{S_r} \varepsilon_0 \vec{E} \, d\vec{s} = Q$$

$$\vec{E} = E_r \vec{e}_r + E_\theta \vec{e}_\theta + E_\phi \vec{e}_\phi$$

~~$E_\theta \vec{e}_\theta$~~ ~~$E_\phi \vec{e}_\phi$~~ sadece yarıçapa bağlı

$$\vec{E} = E_r(r) \vec{e}_r$$

$$d\vec{s} = r^2 \sin\theta \, d\theta \, d\phi \, \vec{e}_r$$

$$\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \varepsilon_0 E_r(r) \vec{e}_r \, r^2 \sin\theta \, d\theta \, d\phi \, \vec{e}_r = Q$$

$$4\pi \varepsilon_0 r^2 E_r(r) = Q \Rightarrow E_r(r) = \frac{Q}{4\pi \varepsilon_0 r^2}$$

$$\boxed{\vec{E} = \frac{Q}{4\pi\varepsilon_0} \frac{\vec{e}_r}{r^2}}$$

$r < a$

$$Q = \int_V \rho_V \, dV$$

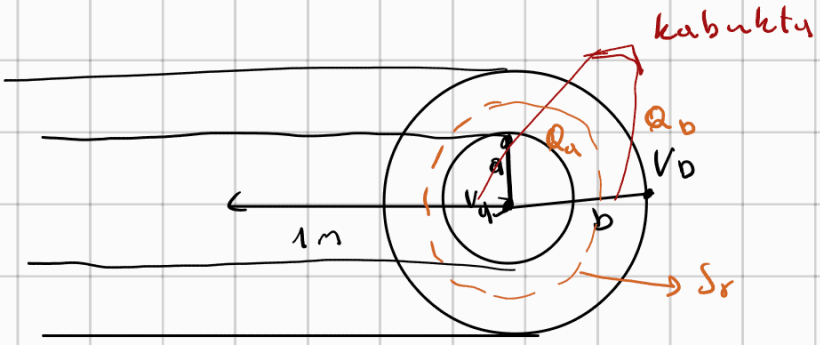
$$= \int_{r'=0}^{r} \int_{\theta'=0}^{\pi} \int_{\phi'=0}^{2\pi} \rho_V \, r^2 \sin\theta \, dr' d\phi' d\theta'$$

$$= \frac{4\pi r^3}{3} \rho_V$$

Sayfa 32 Pr. 2

s26 Pr 5'i gauss ile çözünüz. $V(\infty) = 0$ halinde içteki ve dıştaki silindirin birim yükseklikteki kısmında bulunan yükü V_a ve V_b cinsinden yazınız

iç içe iki silindir



$$\vec{E} = E_\rho \vec{e}_\rho + E_\phi \vec{e}_\phi + E_z \vec{e}_z$$

$$\vec{E} = E_\rho(\rho)\vec{e}_\rho \qquad d\vec{S} = \rho\, d\phi\, dz\, \vec{e}_\rho$$

$a < \rho < b$

$$\int_{S_r} \vec{D}\, d\vec{S} = Q_a \Rightarrow \int_{z=0}^{1}\int_{\phi=0}^{2\pi} \varepsilon_0 E_\rho(\rho)\, \rho\, d\phi\, dz = Q_a \text{ olur}$$

$$2\pi\rho\,\varepsilon_0 E_\rho(\rho) = Q_a \text{ olur}$$

$$\vec{E} = E_\rho(\rho)\vec{e}_\rho = \frac{Q_a}{2\pi\varepsilon_0\rho}\vec{e}_\rho$$

$$\vec{E} = -\text{grad}\, V$$

$$\frac{Q}{2\pi\varepsilon_0\rho}\vec{e}_\rho = -\frac{\partial V}{\partial \rho}\vec{e}_\rho$$

$a < \rho < b$ $\quad V(\rho) = \frac{Q_a}{2\pi\varepsilon_0}\ln\left(\frac{1}{\rho}\right) + C$

$$\rho = a \Rightarrow V_a = \frac{Q_a}{2\pi\varepsilon_0}\ln\left(\frac{1}{a}\right) + C$$

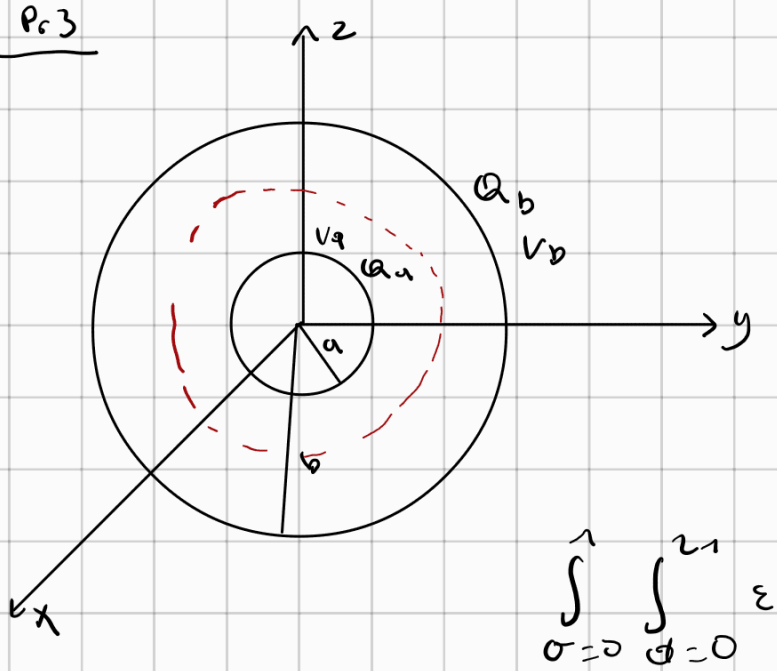
$$\rho = b \Rightarrow V_b = \frac{Q_a}{2\pi\varepsilon_0}\ln\left(\frac{1}{b}\right) + C$$

$\rho > b$ $\quad \int_S \vec{D}\cdot d\vec{S} = Q_a + Q_b$

$$\vec{E} = \frac{Q_a + Q_b}{2\pi\varepsilon_0\rho}\vec{e}_\rho$$

$$V(\rho) = \frac{Q_a + Q_b}{2\pi\varepsilon_0} \ln\left(\frac{1}{\rho}\right) + D$$

S.32 Pr3



ispatla

$$\frac{1}{a} Q_a + \frac{1}{b} Q_b = 4\pi\varepsilon_0 V_0$$

olduğunu gösteriniz

$$a < r < b \quad \int_{S_r} \vec{D}\, d\vec{s} = Q_a$$

$$\vec{E} = E_r(r)\,\vec{e}_r, \qquad d\vec{s} = r^2 \sin\theta\, d\phi\, d\theta\, \vec{e}_r$$

$$\int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \varepsilon_0 E_r r^2 \sin\theta\, d\theta d\phi = Q_a$$

$$4\pi\varepsilon_0 r^2 E_r = Q_a$$

$$E_r = \frac{Q_a \vec{e}_r}{4\pi\varepsilon_0 r^2}, \qquad \vec{E} = -\text{grad}\, V$$

$$\frac{Q_a \vec{e}_r}{4\pi\varepsilon_0 r^2} = -\frac{dV}{dr}\vec{e}_r$$

$$\boxed{V(r) = \frac{Q_a}{4\pi\varepsilon_0 r} + C}$$

$$V_a = \frac{Q_a}{4\pi\varepsilon_0 a} + C$$

$$V_b = \frac{Q_b}{4\pi\varepsilon_0 b} + C$$

$$V_a - V_b = \frac{Q_a}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$$

$$Q_a = \frac{(V_a - V_b)\, 4\pi\varepsilon_0}{\left(\frac{1}{a} - \frac{1}{b}\right)}$$

$r > b$ $\int_{S_r} \vec{D}\, d\vec{S} = Q_a + Q_b$

$$\boxed{\vec{E} = \frac{Q_a + Q_b}{4\pi\varepsilon_0 r^2}\, \vec{e}_r}$$

Burdan $\vec{E} = -\text{grad}\, V \Rightarrow V(r) = \frac{Q_a + Q_b}{4\pi\varepsilon_0 r} + D$

$V(\infty) = 0 \Rightarrow D = 0$

$$V_b = \frac{Q_a + Q_b}{4\pi\varepsilon_0 b}$$

$$V_a - V_b = \frac{Q_a}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$$

+

$$4\pi\varepsilon_0 V_a = \frac{Q_a}{b} + \frac{Q_b}{b} + \frac{Q_a}{a} - \frac{Q_a}{b}$$

$$\left(\frac{Q_a}{a} + \frac{Q_b}{b} = 4\pi\varepsilon_0 V_a\right)$$

Önemli yerler

Kolomb kanunu

süperposition

Enerji

$\Delta V = 0$ sorusu gelebilir

$\Delta V = -\frac{\rho}{\varepsilon_0}$: Poisson denklemi